Kernel Methods

Learning Theory 7

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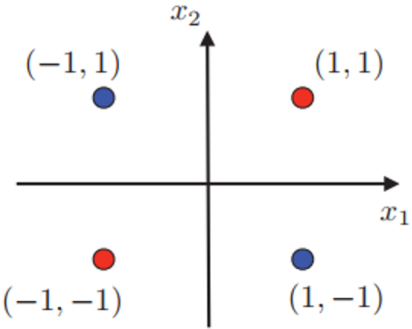
XOR

- The label is **True** if and only if two inputs are not equal.
- The label is **False** if and only if two inputs are equal.
- It is not linear separable.



XOR

- If we define a feature mapping $\Phi : \mathbb{R}^2 \to \mathbb{R}^6$:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \\ \sqrt{2c}x_1 \\ \sqrt{2c}x_2 \\ c \end{pmatrix},$$

where $c > 0$ is an arbitrary constant,

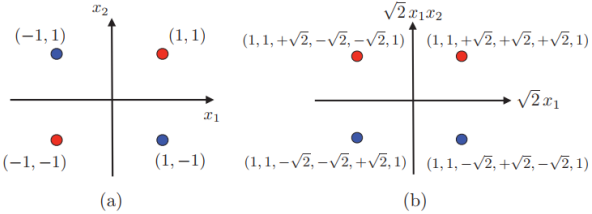
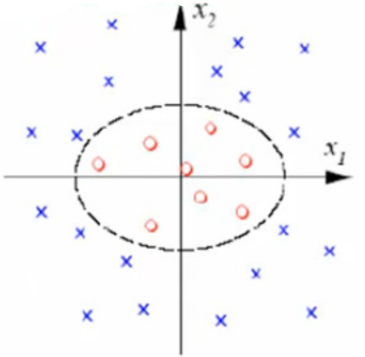


Figure 1: We show the 2-dimensional space defined by the 3rd and 4th dim of $\mathbb{R}^6$, here $c = 1$

- Then it is linear separable in $\mathbb{R}^6$ space.
- (This is a second-degree polynomial kernel, which will be discussed later.)

SVM

- Assume the margin is $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$
- It is not linear separable.



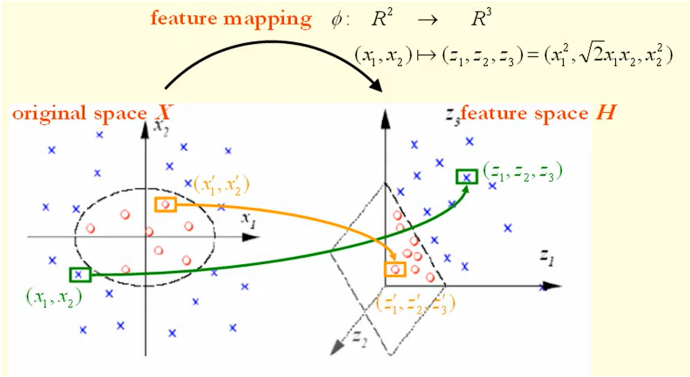
SVM

- If we define a feature mapping:

$$\Phi : \mathbb{R}^2 \to \mathbb{R}^3$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{pmatrix}$$

- Then it is linear separable in $\mathbb{R}^3$ space.



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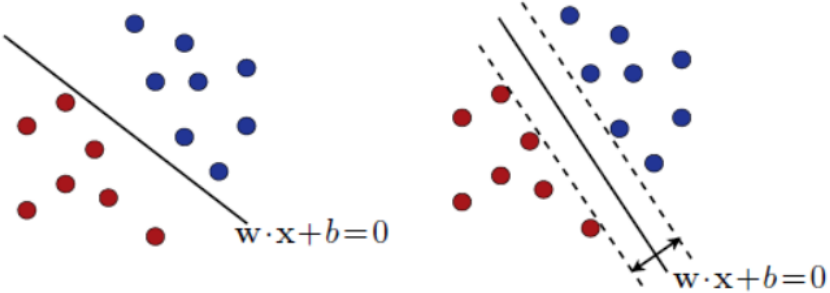
SVM Review



- SVM solution: the separating hyperplane with the maximum geometric margin
- The maximum margin $\rho = \max_{w,b} \min_{i \in [m]} \frac{y_i(w^T x_i + b)}{||w||}$

SVM Review

- The equivalent Primal optimization problem:

$$\min_{w,b} \frac{1}{2}||w||^2, y_i(w^T x_i + b) \geq 1, \forall i \in [m] \tag{1}$$

- The KKT conditions:

$$\begin{aligned} w = \sum_{i=1}^{m} \alpha_i y_i x_i, \sum_{i=1}^{m} \alpha_i y_i = 0 \\ \alpha_i[y_i(w^T x_i + b) - 1] = 0, \forall i \in [m] \end{aligned} \tag{2}$$

- The Dual problem:

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{m} \alpha_i \alpha_j y_i y_j x_i^T x_j \\ s.t. \quad & \sum_{i=1}^{m} \alpha_i y_i = 0, \alpha_i \geq 0, \forall i \in [m] \end{aligned} \tag{3}$$

SVM Review

- The Dual problem:

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{m} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \\ s.t. \quad & \sum_{i=1}^{m} \alpha_i y_i = 0, \alpha_i \geq 0, \forall i \in [m] \end{aligned} \tag{4}$$

- Calculate $w^T x$:

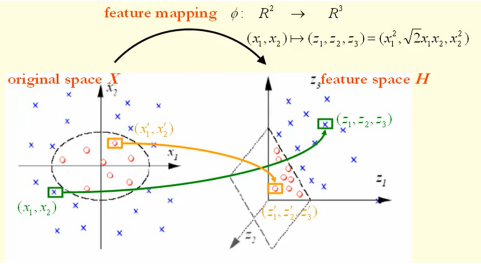
$$w^T x = \sum_{i=1}^{m} \alpha_i y_i \langle x_i, x \rangle \tag{5}$$

- The hypothesis solution only depends on **inner products** between vectors.

Inner Product

- Calculate the inner product in **feature space**:

$$
\begin{aligned}
&\langle (z_1, z_2, z_3), (z_1', z_2', z_3') \rangle \\
=&\langle \Phi(x_1, x_2), \Phi(x_1', x_2') \rangle \\
=&\langle (x_1^2, \sqrt{2}x_1x_2, x_2^2), (x_1'^2, \sqrt{2}x_1'x_2', x_2'^2) \rangle \\
=&(x_1x_1' + x_2x_2')^2 \\
=&(\langle x, x' \rangle)^2 \\
=&\mathcal{K}(x, x')
\end{aligned}
\tag{6}
$$



- The function $\mathcal{K}$ is the so called **kernel**.

Kernel Function

Definition (Kernel)

A function $\mathcal{K} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is called a kernel over $\mathcal{X}$.

- The idea is to define a kernel $\mathcal{K}$ such that

$$\forall x, x' \in \mathcal{X}, \mathcal{K}(x, x') = \langle \Phi(x), \Phi(x') \rangle \tag{7}$$

- Kernel function $\longrightarrow$ Geometric properties in feature space
- Distance in $\mathcal{H}$:

$$\begin{aligned} ||\Phi(x) - \phi(x')||^2 &= \langle \Phi(x), \Phi(x) \rangle - 2\langle \Phi(x), \Phi(x') + \langle \Phi(x'), \Phi(x') \rangle \\ &= \mathcal{K}(x, x) - 2\mathcal{K}(x, x') + \mathcal{K}(x', x') \end{aligned}$$

- Angle in $\mathcal{H}$: $\cos\theta = \frac{\langle \Phi(x'), \Phi(x') \rangle}{||\Phi(x)|| \cdot ||\Phi(x')||} = \frac{\mathcal{K}(x,x')}{\sqrt{\mathcal{K}(x,x)}\sqrt{\mathcal{K}(x',x')}}$

Kernel & Feature Mapping

- Is Φ necessary? Can we only employ $\mathcal{K}$?

$$\rightarrow \text{Yes!}$$

- Given a kernel function $\mathcal{K}$, can we construct a feature space $\mathcal{H}$(i.e., a function mapping Φ) s.t. $\mathcal{K}$ computes the inner product in $\mathcal{H}$?

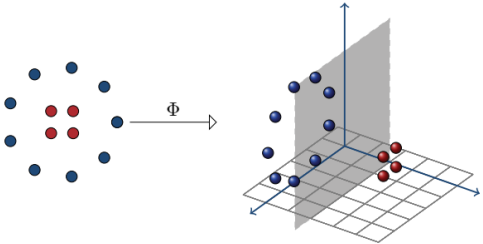
$$\rightarrow \text{Yes!}$$

- What kind of $\mathcal{K}$ can be used?

$$\rightarrow \text{PSD (Positive semidefinite) kernel}$$

Kernel Methods

- Kernel methods work by
  - Embedding data in a vector space (with higher dimensionality)
  - Looking for (linear) relations in such space
- If the map Φ is chosen suitably, complex relations can be simplified, and easily detected.



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PSD Kernel: Definition

Definition (Positive semidefinite kernel function)

A symmetric bivariate function $\mathcal{K} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is positive semidefinite (PSD) if **for all integers** $n \geq 1$ **and elements** $\{x_i\}_{i=1}^n \subset \mathcal{X}$, the $n \times n$ matrix with elements $\mathbf{K}_{ij} := \mathcal{K}(x_i, x_j)$ is positive semidefinite.

For a sample $S = (x_1, \ldots, x_n)$, $\mathbf{K}$ is called the **kernel matrix** or the **Gram matrix** associated to $\mathcal{K}$ and the sample S.

PSD Kernel: Example

Polynomial kernels

$$\mathcal{K}(x, x') = (x \cdot x' + c)^d$$

where $c > 0$ is a constant, $d \in \mathbb{N}$ is the degree of polynomial kernel

- Polynomial kernels map the **input space** to a **higher-dimensional space** of dimension $\binom{N+d}{d}$, where N is the dim of input.
- When d = 2, N = 2, the dim of the feature space is 6:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \\ \sqrt{2c}x_1 \\ \sqrt{2c}x_2 \\ c \end{pmatrix}$$

PSD Kernel: Example

Theorem

Polynomial kernels map the input space to a higher-dimensional space of dimension $\binom{N+d}{d}$*, where* d *is the degree of polynomial kernel,* N *is the dim of input.*

Proof.

Denote $y_i = x_i x_i^T$, then $\mathcal{K}(x, x') = (\sum_{i=1}^N y_i + c)^d$. Consider the set of basis $\mathcal{B}_n = \{\prod_{i=1}^N y_i^{d_i} | \sum_{i=1}^N d_i \leq d, d_i \in \mathbb{N}\}$, where the elements of the set is the number of solutions of the inequality $\sum_{i=1}^N d_i \leq d$. We add a slack variable $\xi \in \mathbb{N}$, then it is transferred to an indefinite equation $\sum_{i=1}^N d_i + \xi = d$. Equivalently, the sum of $N+1$ variables is d. From the plug-in method, the number of solutions to this equation is

$$\binom{d + (N+1) - 1}{(N+1) - 1} = \binom{d+N}{N} = \binom{d+N}{d}$$

PSD Kernel: Example

Polynomial kernels $\mathcal{K}(x, x') = (x \cdot x' + c)^d$

Prove that $\mathcal{K}(x, z) = \langle x, z \rangle + 1$ is PSD Kernel ($d = 1, c = 1$):
For any column vector α,

$$\alpha^T \mathcal{K} \alpha = \alpha^T \begin{pmatrix} x_1^T x_1 & \cdots & x_1^T x_n \\ \vdots & \vdots & \vdots \\ x_n^T x_1 & \cdots & x_n^T x_n \end{pmatrix} \alpha + \alpha^T \alpha = (\begin{pmatrix} x_1 & \cdots & x_n \end{pmatrix} \alpha)^T (\begin{pmatrix} x_1 & \cdots & x_n \end{pmatrix} \alpha) + \alpha^T \alpha \geq 0$$

PSD Kernel: Example

Gaussian Kernels (RBF, Radial Basis Function Kernel)

$$\mathcal{K}(x, x') = \exp(-\frac{||x' - x||^2}{2\sigma^2})$$

where $\sigma > 0$ is a constant

- The feature space has an infinite number of dimensions. For $\sigma = 1$,

$$\exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{x}'\|^2\right) = \sum_{j=0}^{\infty} \sum_{\sum n_i = j} \exp\left(-\frac{1}{2}\|\mathbf{x}\|^2\right) \frac{x_1^{n_1} \cdots x_k^{n_k}}{\sqrt{n_1! \cdots n_k!}} \exp\left(-\frac{1}{2}\|\mathbf{x}'\|^2\right) \frac{x_1'^{n_1} \cdots x_k'^{n_k}}{\sqrt{n_1! \cdots n_k!}}$$

- The Gaussian Kernel is **universal**.[1]

[1] For interested readers, please see *Micchelli, C. A., Xu, Y., & Zhang, H. (2006). Universal Kernels. Journal of Machine Learning Research, 7(12).*

PSD Kernel: Example

Sigmoid Kernels

$$\mathcal{K}(x, x') = \tanh(a(x \cdot x') + b)$$

where $a, b \geq 0$ are constants, $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Exponential Kernels

$$\mathcal{K}(x, x') = \exp(-\frac{||x' - x||}{2\sigma^2})$$

where $\sigma > 0$ is a constant

Laplacian Kernels

$$\mathcal{K}(x, x') = \exp(-\frac{||x' - x||}{\sigma})$$

where $\sigma > 0$ is a constant

PSD Kernel: Example

Note: Some papers say that ***The Laplace Kernel is completely equivalent to the exponential kernel, except for being less sensitive for changes in σ parameter.***, such as *Fadel, S., Ghoniemy, S., Abdallah, M., Sorra, H. A., Ashour, A., & Ansary, A. (2016). Investigating the effect of different kernel functions on the performance of SVM for recognizing Arabic characters. International Journal of Advanced Computer Science and Applications, 7(1), 446-450.*, *Ali, Z., Shahzad, S. K., & Shahzad, W. (2018). Performance analysis of support vector machine based classifiers. International Journal of Advanced and Applied Sciences, 5(9), 33-38.*, and they are types of RBF. Other papers say that **exponential kernel is** $\mathcal{K}(x, y) = \exp(-\lambda(d(x, y))^q), \lambda, q > 0$**, where** $d(x, y)$ **is geodesic distance. For** $q = 2$ **this gives a geodesic generalization of the Gaussian kernel, and** $q = 1$ **gives the geodesic Laplacian kernel.** For example, *Feragen, A., Lauze, F., & Hauberg, S. (2015). Geodesic exponential kernels: When curvature and linearity conflict. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 3032-3042)*.

Hilbert Space

Definition (Hilbert Space)

A Hilbert space is a **vector space** equipped with an **inner product** which defines a **distance function** for which it is a **complete metric space**.

Hilbert Space

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A Hilbert space is a **vector space** equipped with an **inner product** which defines a **distance function** for which it is a **complete metric space**.

Vector space

For any $u, v, w \in V$, and a, b in $\mathcal{F}$,

1. $u + (v + w) = (u + v) + w$
2. $u + v = v + u$
3. $\exists 0 \in V, v + 0 = v$
4. $\exists -v \in V, v + (-v) = 0$
5. $a(bv) = (ab)v$
6. $1v = v$
7. $a(u + v) = au + av$
8. $(a + b)v = av + bv$

Hilbert Space

Definition (Hilbert Space)

A Hilbert space is a **vector space** equipped with an **inner product** which defines a **distance function** for which it is a **complete metric space**.

Inner product

For a real vector space, an inner product $\langle \cdot, \cdot \rangle$ satisfies the following four properties. Let u, v, and w be vectors and α be a scalar, then:

- $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
- $\langle \alpha v, w \rangle = \alpha, w \rangle$
- $\langle v, w \rangle = \langle w, v \rangle$
- $\langle v, v \rangle = 0$ and equal if and only if $v = 0$

Then we can define a norm $||x|| = \langle x, x \rangle$, and **distance function** $d(x, y) = ||x - y||$.

Hilbert Space

Definition (Hilbert Space)

A Hilbert space is a **vector space** equipped with an **inner product** which defines a **distance function** for which it is a **complete metric space**.

Complete metric space

A metric space is complete if every Cauchy sequence of points in it has a limit that is also in it.

Cauchy sequence

A sequence $\{x_1, x_2, \ldots\}$ is called Cauchy sequence if $\forall r > 0, \exists N \in \mathbb{N}$ s.t. $\forall m, n > N, d(x_m, x_n) < r$.

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RKHS, Reproducing Kernel Hilbert Space

Definition (RKHS, Reproducing Kernel Hilbert Space)

Let $\mathcal{X}$ be a nonempty set (often called the index set) and $\mathcal{H}$ a Hilbert space of functions $f : \mathcal{X} \to \mathbb{R}$. Then $\mathcal{H}$ is called a **reproducing kernel Hilbert space** endowed with the inner product $\langle \cdot, \cdot \rangle$ (and the norm $||f|| := \sqrt{\langle f, f \rangle}$)if there exists a function $\mathcal{K} : \mathcal{X} \times \mathcal{X}$ with the **reproducing property**:

$$\langle f, \mathcal{K}(\cdot, x) \rangle = f(x), \forall f \in \mathcal{H}; \tag{8}$$

in particular,

$$\langle \mathcal{K}(\cdot, x), \mathcal{K}(\cdot, z) \rangle = \mathcal{K}(x, z) \tag{9}$$

RKHS

Question: How to build RKHS when given a PSD kernel $\mathcal{K}$?
We have 2 approaches:

- Approach 1: By applying Mercer's Theorem
- Approach 2: By building the Hilbert space step by step

Mercer's Theorem

Theorem (Mercer's Theorem)

Suppose that X is compact, the kernel function $\mathcal{K}$ is continuous and positive semidefinite, and satisfies the Hilbert-Schmidt condition

$$\int_{\mathcal{X}\times\mathcal{X}} K^2(x,z)d\mathbb{P}(x)d\mathbb{P}(z) < \infty \tag{10}$$

Then there exist a sequence of eigenfunctions $(\phi_j)_{j=1}^{\infty}$ that form an orthonormal basis of $L^2(\mathcal{X};\mathbb{P})$(with the usual squared norm $||f||^2_{L^2(\mathcal{X};\mathbb{P})} = \int_{\mathcal{X}} f^2(x)d\mathbb{P}(x)$), and non-negative eigenvalues $(\mu_j)_{j=1}^{\infty}$ such that

$$T_{\mathcal{K}}(\phi_j) = \mu_j\phi_j, \forall j = 1, 2, \ldots \tag{11}$$

where $T_{\mathcal{K}}$ is a linear operator on $L^2(\mathcal{X})$ via $T_{\mathcal{K}}(f)(x) := \int_{\mathcal{X}} \mathcal{K}(x,z)f(z)d\mathbb{P}(z)$. Moreover, the kernel function has the expansion

$$\mathcal{K}(x,z) = \sum_{j=1}^{\infty} \mu_j\phi_j(x)\phi_j(z) \tag{12}$$

where the convergence of the infinite series holds absolutely and uniformly.

From Mercer's Theorem to RKHS

Given the eigenfunctions and eigenvalues guaranteed by Mercer's theorem, we may define a mapping $\Phi : \mathcal{X} \to \mathcal{H}$ via

$$x \mapsto \Phi(x) := (\sqrt{\mu_1}\phi_1(x), \sqrt{\mu_2}\phi_2(x), \sqrt{\mu_3}\phi_3(x), \cdots)$$

This feature map provides an explicit inner product representation of the kernel over $\mathcal{H}$, namely

$$\langle \Phi(x), \Phi(z) \rangle_{\mathcal{H}} = \sum_{j=1}^{\infty} \mu_j \phi_j(x)\phi_j(z) = \mathcal{K}(x, z)$$

From Mercer's Theorem to RKHS

Corollary Consider a kernel satisfying the conditions of Mercer's theorem with associated eigenfunctions $(\phi_j)_{j=1}^{\infty}$ and non-negative eigenvalues $(\mu_j)_{j=1}^{\infty}$. It induces the reproducing kernel Hilbert space

$$\mathcal{H} := \left\{ f = \sum_{j=1}^{\infty} \beta_j \phi_j \mid \text{ for some } (\beta_j)_{j=1}^{\infty} \in \ell^2(\mathbb{N}) \text{ with } \sum_{j=1}^{\infty} \frac{\beta_j^2}{\mu_j} < \infty \right\} \tag{13}$$

along with inner product

$$\langle f, g \rangle_{\mathcal{H}} := \sum_{j=1}^{\infty} \frac{\langle f, \phi_j \rangle \langle g, \phi_j \rangle}{\mu_j} \tag{14}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product in $L^2(\mathcal{X}; \mathbb{P})$.

From Mercers Theorem to RKHS

Proof of reproducing property

- By Mercer expansion, we have $\mathcal{K}(\cdot, x) = \sum_{j=1}^{\infty} \mu_j \phi_i(x) \phi_j(\cdot)$.
- By the orthonormality of $(\phi_j)_{j=1}^{\infty}$ in $L^2(\mathcal{X}; \mathbb{P})$ and Mercer's theorem, we have $\langle \mathcal{K}(\cdot, x), \phi_j \rangle = \mu_j \phi_j(x)$ for each $j \in \mathbb{N}$.
- By definition (14) of Hilbert inner product, for any $f \in \mathcal{H}$, we have

$$\langle f, \mathcal{K}(\cdot, x) \rangle_{\mathcal{H}} = \sum_{j=1}^{\infty} \frac{\langle f, \phi_j \rangle \langle \mathcal{K}(\cdot, x), \phi_j \rangle}{\mu_j} = \sum_{j=1}^{\infty} \langle f, \phi_j \rangle \phi_j(x) = f(x),$$

where the final step again uses the orthonormality of $(\phi_j)_{j=1}^{\infty}$.

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Construction schedule

Another Approach: Construction schedule

- Step 1: Define a vector space based on the image of Φ (Φ is called kernel map)
- Step 2: Define an inner product
- Step 3: Show that $\mathcal{K}(x, z) = \langle \Phi(x), \Phi(z) \rangle$

Step 1: Define a vector space based on the image of Φ

Definition (Reproducing Kernel Map)

Suppose $\mathcal{K} = \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is a kernel. The corresponding **kernel map** is a map

$$\Phi : \mathbb{R}^d \to \mathbb{R}^{\mathbb{R}^d} \text{ such that } \Phi(z) = \varphi_z = \mathcal{K}(\cdot, z)$$

that is,

$$\Phi(z)(x) = \varphi_z(x) = \mathcal{K}(x, z)$$

Example (Gaussian Kernel)

Given $\mathcal{K}(x, z) = \exp(-\frac{(x-z)^2}{2})$

If $z = 0$, $\Phi(0)(x) = \varphi_0(x) = \mathcal{K}(x, 0) = \exp(-\frac{x^2}{2})$

If $z = 1$,$\Phi(1)(x) = \varphi_1(x) = \mathcal{K}(x, 1) = \exp(-\frac{(x-1)^2}{2})$

Step 1: Define a vector space based on the image of Φ

Suppose $\mathcal{K} : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is a kernel and Φ is the corresponding reproducing kernel map. Let

$$\mathcal{F}_{\mathcal{K}} = \{f = \sum_{i=1}^{m} \alpha_i \Phi(x_i) | m \in \mathbb{N}, x_1, \cdots, x_m \in \mathbb{R}^d, \alpha_1, \cdots, \alpha_m \in \mathbb{R}\}$$

Then $\mathcal{F}_{\mathcal{K}}$ is a vector space over $\mathbb{R}$.

Example (Gaussian Kernel)

Given $\mathcal{K}(x, z) = \exp(-\frac{(x-z)^2}{2})$

$\Phi(0)(x) = \varphi_0(x) = \mathcal{K}(x, 0) = \exp(-\frac{x^2}{2})$

$\Phi(1)(x) = \varphi_1(x) = \mathcal{K}(x, 1) = \exp(-\frac{(x-1)^2}{2})$

$g(x) = -1 \cdot \Phi(0)(x) + 1 \cdot \Phi(1)(x)$

$= -\exp(-\frac{x^2}{2}) + \exp(-\frac{(x-1)^2}{2}) \in \mathcal{F}_{\mathcal{K}}$

Step 2: Define an inner product

Given $f = \sum_{i=1}^{m} \alpha_i \Phi(x_i)$ and $g = \sum_{j=1}^{m'} \beta_j \Phi(x'_j)$

Recall that $f = \sum_{i=1}^{m} \alpha_i \mathcal{K}(\cdot, x_i)$ and $g = \sum_{j=1}^{m'} \beta_j \mathcal{K}(\cdot, x_i)$

If we define $\langle f, g \rangle := \sum_{i=1}^{m} \sum_{j=1}^{m'} \alpha_i \beta_j \mathcal{K}(x_i, x'_j)$, then it can be shown that $\langle f, g \rangle$ is an inner product over $\mathcal{F}_K$

Step 2: Define an inner product

Show that $\langle f, g\rangle = \sum_{i=1}^{m}\sum_{j=1}^{m'} \alpha_i\beta_j\mathcal{K}(x_i, x_j')$ is an inner product over $\mathcal{F}_K$.

- Symmetry: $\langle f, g\rangle = \langle g, f\rangle$ Obviously.
- Linearity: $\langle af, g\rangle = a\langle f, g\rangle, \langle f + h, g\rangle = \langle f, g\rangle + \langle h, g\rangle$ Obviously.
- Positive-defiteness: $\langle f, f\rangle = \sum_{i=1}^{m}\sum_{j=1}^{m} \alpha_i\alpha_j\mathcal{K}(x_i, x_j) \geq 0$, since K is PSD. $|f(x)|^2 = |\langle\mathcal{K}(\cdot, x), f\rangle|^2 \leq \langle\mathcal{K}(\cdot, x), \mathcal{K}(\cdot, x)\rangle\langle f, f\rangle = \mathcal{K}(x, x)\langle f, f\rangle$. Hence $\langle f, f\rangle = 0$ implies $|f(x)| = 0, \forall x$, then $f(x) = 0, \forall x$, $f = 0$. The reverse is easy.

Step 2: Define an inner product

Example

Given $\mathcal{K}(x,z) = \exp(-\frac{(x-z)^2}{2})$

$f(x) = \mathcal{K}(x,0) - \mathcal{K}(x,1) = \exp(-\frac{x^2}{2}) - \exp(-\frac{(x-1)^2}{2})$

$g(x) = -\mathcal{K}(x,0) + \mathcal{K}(x,1) = -\exp(-\frac{x^2}{2}) + \exp(-\frac{(x-1)^2}{2})$

$\langle f, g\rangle = 1\cdot(-1)\mathcal{K}(0,0) + 1\cdot 1\mathcal{K}(0,1) + (-1)\cdot(-1)\mathcal{K}(1,0) + (-1)\cdot 1\mathcal{K}(1,1)$

$\langle f, g\rangle = 1\cdot g(0) + (-1)\cdot g(1)$

$\langle f, g\rangle = (-1)\cdot f(0) + 1\cdot f(1)$

Step 2: Define an inner product

$$\langle f, g\rangle = \sum_{i=1}^{m}\sum_{j=1}^{m'} \alpha_i\beta_j\mathcal{K}(x_i, x_j') = \sum_{i=1}^{m}\alpha_i \sum_{j=1}^{m'}\beta_j\mathcal{K}(x_i, x_j') = \sum_{i=1}^{m}\alpha_i g(x_i) \tag{15}$$

Since $g = \sum_{j=1}^{m'}\beta_j\mathcal{K}(\cdot, x_j')$. Similarly,

$$\langle f, g\rangle = \sum_{i=1}^{m}\sum_{j=1}^{m'} \alpha_i\beta_j\mathcal{K}(x_i, x_j') = \sum_{j=1}^{m'}\beta_j \sum_{i=1}^{m}\alpha_i\mathcal{K}(x_j', x_i) = \sum_{i=1}^{m'}\beta_j f(x_j') \tag{16}$$

Since $f = \sum_{i=1}^{m}\alpha_i\mathcal{K}(\cdot, x_i)$.

Proposition: Representer of evaluation

- $\langle \mathcal{K}(\cdot, x), f\rangle = f(x) = \sum_{i=1}^{m}\alpha_i\mathcal{K}(x_i, x)$
- $\langle \mathcal{K}(\cdot, x), \mathcal{K}(\cdot, z)\rangle = \mathcal{K}(z, x) = \mathcal{K}(x, z)$

Step 3: Show that $\mathcal{K}(x, z) = \langle \Phi(x), \Phi(z) \rangle$

$$\mathcal{K}(x, z) = \langle \mathcal{K}(\cdot, x), \mathcal{K}(\cdot, z) \rangle = \langle \Phi(x), \Phi(z) \rangle$$

since $\Phi(x) = \varphi_x = \mathcal{K}(\cdot, x)$, and $\Phi(z) = \varphi_z = \mathcal{K}(\cdot, z)$

We now have shown that any PSD kernel can be thought of as a inner product in another space.

1 Motivating Examples

2 Kernel Methods

3 PSD Kernel

4 RKHS and Mercer's Theorem

5 RKHS: Another Approach

6 Statistical Learning

Representer Theorem

Theorem (Representer Theorem)

Consider a regularized risk minimization problem of the form

$$\underset{\theta \in \mathcal{H}}{\text{minimize}} \quad \frac{1}{n} \sum_{i=1}^{n} l(y_i, \langle \theta, \Phi(x_i) \rangle) + \lambda ||\theta||_{\mathcal{H}}^2 \tag{17}$$

where $\mathcal{X}$ arbitrary input space, $\mathcal{Y}$ output space, $\mathcal{K} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ a kernel, $\mathcal{H}$ the corresponding RKHS, l the loss function. [2] *Problem (17) always has an optimal solution of the form*

$$w^* = \sum_{i=1}^{n} \alpha_i \mathcal{K}(\cdot, X_i) \tag{18}$$

[2] Note that there is no assumption on the loss function l, in particular no convexity is assumed.

Representer Theorem

Proof.

1. Split a the space $\mathcal{H}$ into the subspace $\mathcal{H}_{data} := span\{K_{X_1}, \cdots, K_{X_n}\}$ (induced by the data) and its orthogonal complement $\mathcal{H}_{comp}$. Then $\mathcal{H} = \mathcal{H}_{data} + \mathcal{H}_{comp}$.
2. Now express each vector $w \in \mathcal{H}$ as $w = w_{data} + w_{comp}$.
3. It is not difficult to see that the predictions of all functions with the same w_{data} agree on all training points, they do not depend on w_{comp}.
4. So in particular, the loss w is not affected by w_{comp}.
5. For fixed w_{data}, the norm of w is smallest if w_{comp} is 0.
6. So if we had a solution w^* where w_{comp} would be non-zero, we could get a better solution by setting $w_{comp} = 0$.
7. Thus we can always find an optimal solution with $w_{comp} = 0$.

Representer Theorem

Corollary

Problem (17) can be reformulated as

$$\underset{\alpha \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{n} \sum_{i=1}^{n} l(y_i, (\mathbf{K}\alpha)_i) + \lambda \alpha^T \mathbf{K} \alpha \tag{19}$$

Proof.

Please see Bach Corollary 7.1. □

Kernel ridge regression

Example (Kernel ridge regression)

Recall ridge regression in feature space:

$$\text{minimize}_w \quad \frac{1}{n}\sum_{i=1}^{n}(Y_i - \langle w, \Phi(X_i)\rangle)^2 + \lambda||w||^2 \tag{20}$$

Apply the representer theorem, we express w as a linear combination of input points:

$$w = \sum_{j=1}^{n} \alpha_j \Phi(X_j) \tag{21}$$

Kernel ridge regression

Example (Kernel ridge regression)

This leads to the following kernelized ridge regression problem:

$$\text{minimize}_{\alpha} \quad \frac{1}{n}||Y - \mathbf{K}\alpha||_2^2 + \lambda\alpha^T\mathbf{K}\alpha \tag{22}$$

The solution is given by

$$\alpha = (n\lambda I + \mathbf{K})^{-1}Y \tag{23}$$

Then, we can compute the prediction for a new test point just using kernels.

Kernel SVM

Example (Kernel SVM)

Kernelized dual SVM problem:

$$\begin{array}{ll} \underset{\alpha \in \mathbb{R}^n}{\text{maximize}} & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j Y_i Y_j \mathcal{K}(X_i, X_j) \\ \text{subject to} & \alpha_i \geq 0 \quad \forall i = 1, \ldots, n \\ & \sum_{i=1}^n \alpha_i Y_i = 0 \end{array}$$

Solving this problem gives the dual variables α.

Kernel SVM

Example (Kernel SVM)

Computing labels at new points:
Then, as in SVM :

- $w = \sum_i \alpha_i Y_i X_i$
- $b = Y_j - \sum_i Y_i \alpha_i \langle X_i, X_j \rangle$ for some j such that $\alpha_j > 0$
- Label of test point X given by $\langle w, X \rangle + b$

In kernel language:

$$\langle w, X \rangle + b = \left\langle \sum_i \alpha_i Y_i X_i, X \right\rangle + b = \sum_i \alpha_i Y_i \mathcal{K}(X_i, X) + \left(Y_j - \sum_i Y_i \alpha_i \mathcal{K}(X_i, X_j) \right)$$

Rademacher Complexity

Theorem (Rademacher complexity of kernel-based hypotheses)

Let $\mathcal{K} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a PSD kernel and let $\Phi : \mathcal{X} \to \mathcal{H}$ be a feature mapping associated to K. Let $S \subseteq \{x : \mathcal{K}(x, x) \leq r^2\}$ be a sample of size m, and let $H = \{\mathbf{x} \mapsto \langle \mathbf{w}, \Phi(x) \rangle : \|\mathbf{w}\|_{\mathcal{H}} \leq \Lambda\}$ for some $\Lambda \geq 0$. Then

$$\widehat{\mathfrak{R}}_S(H) \leq \frac{\Lambda\sqrt{\mathrm{Tr}[\mathbf{K}]}}{m} \leq \sqrt{\frac{r^2\Lambda^2}{m}} \tag{24}$$

where $\widehat{\mathfrak{R}}_S(H)$ is the empirical Rademacher complexity of H.

Rademacher Complexity

Proof.

$$
\begin{aligned}
\widehat{\mathfrak{R}}_S(H) &= \frac{1}{m}\mathop{\mathrm{E}}_{\boldsymbol{\sigma}}\left[\sup_{\|\mathbf{w}\|\le\Lambda}\left\langle \mathbf{w}, \sum_{i=1}^m \sigma_i \Phi(x_i)\right\rangle\right] \\
&= \frac{\Lambda}{m}\mathop{\mathrm{E}}_{\boldsymbol{\sigma}}\left[\left\|\sum_{i=1}^m \sigma_i \Phi(x_i)\right\|_{\mathcal{H}}\right] \qquad \text{(Cauchy-Schwarz, eq. case)} \\
&\le \frac{\Lambda}{m}\left[\mathop{\mathrm{E}}_{\boldsymbol{\sigma}}\left[\left\|\sum_{i=1}^m \sigma_i \Phi(x_i)\right\|_{\mathcal{H}}^2\right]\right]^{1/2} \qquad \text{(Jensen's ineq.)} \\
&= \frac{\Lambda}{m}\left[\mathop{\mathrm{E}}_{\boldsymbol{\sigma}}\left[\sum_{i=1}^m \|\Phi(x_i)\|_{\mathcal{H}}^2\right]\right]^{1/2} \qquad \left(i \ne j \Rightarrow \mathop{\mathrm{E}}_{\boldsymbol{\sigma}}[\sigma_i\sigma_j] = 0\right) \\
&= \frac{\Lambda}{m}\left[\mathop{\mathrm{E}}_{\boldsymbol{\sigma}}\left[\sum_{i=1}^m K(x_i, x_i)\right]\right]^{1/2} = \frac{\Lambda\sqrt{\mathrm{Tr}[\mathbf{K}]}}{m} \le \sqrt{\frac{r^2\Lambda^2}{m}}.
\end{aligned}
$$

Rademacher Complexity

Corollary (Generalization bound)

Let $\mathcal{K} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a PSD kernel with $r = \sup_{x \in \mathcal{X}} \mathcal{K}(x, x)$. Let $\Phi : \mathcal{X} \to \mathcal{H}$ be a feature mapping associated to $\mathcal{K}$ and let $H = \{\mathbf{x} \mapsto \mathbf{w} \cdot \Phi(x) : \|\mathbf{w}\|_{\mathcal{H}} \leq \Lambda\}$ for some $\Lambda \geq 0$. Then, for any $\delta > 0$, each of the following statements holds with probability at least $1 - \delta$ for any $h \in H$:

$$R(h) \leq \widehat{R}(h) + 2\sqrt{\frac{r^2\Lambda^2/\rho^2}{m}} + \sqrt{\frac{\log \frac{1}{\delta}}{2m}}$$

$$R(h) \leq \widehat{R}(h) + 2\frac{\sqrt{\mathrm{Tr}[\mathbf{K}]\Lambda^2/\rho^2}}{m} + 3\sqrt{\frac{\log \frac{2}{\delta}}{2m}}$$

Contents we haven't covered

- Kosambi-Karhunen-Loève theorem (KL representer theorem). Link: `https://en.wikipedia.org/wiki/Kosambi%E2%80%93Karhunen%E2%80%93Lo%C3%A8ve_theorem`
- Theoretical analysis of ridge regression: Please reference Bach Section 7.6.

References

- Bach, Chapter 7
- Wainwright, Chapter 12
- Understanding Machine Learning, Chapter 16
- Foundations of Machine Learning, Chapter 5(1st ed.) & Chapter 6(2nd ed.)
- The Elements of Statistical Learning, Chapter 5.8. Here's illustration of eigenvectors of gaussian kernel matrix $\mathbf{K}$.
- Course *Kernel Method* by Cheng-Hsuan Li. `https://www.youtube.com/watch?v=p4t6O9uRX-U&list=PLt0SBi1p7xrRKE2us8doqryRou6eDYEOy`
- Statistical Machine Learning slides by Ulrike von Luxburg. `https://www.tml.cs.uni-tuebingen.de/teaching/2020_statistical_learning/downloads_free/luxburg_statistical_learning_slides.pdf`

Thanks!